132 Pattern (medium)

Problem Statement

Given an array `nums`, containing `N` integers.

A `132 pattern` consists of three numbers, say x, y, and z, where `x < z` and `z < y`. This is often referred to as a '132' pattern because if we represent x, y, and z as 1, 3, and 2, respectively, it mimics the positional pattern in '132'.

Return `true` if such a pattern exists within a given sequence of numbers. Otherwise, return `false`.

Examples

1. **Example 1**:
   - Input: nums = [3, 5, 0, 3, 4]
   - Expected Output: True
   - Justification: Here, 3 < 4 and 4 < 5, forming a '132' pattern with the numbers 3, 5, and 4.
2. **Example 2**:
   - Input: nums = [1, 2, 3, 4]
   - Expected Output: False
   - Justification: The sequence is in ascending order, and no '132' pattern is present.
3. **Example 3**:
   - Input: nums = [9, 11, 8, 9, 10, 7, 9]
   - Expected Output: True
   - Justification: The pattern is formed with 9 < 10 and 10 < 11.

Try it yourself

Try solving this question here:

Java

```java
import java.util.TreeSet;

public class Solution {
    // Method to check if the array contains a 132 pattern
    public boolean find132pattern(int[] nums) {
        // ToDo: Write Your Code Here.
        return false;
    }
}
```

Test Case | Run Code Result | Stdout

[1, 2, 3, 4]

Use Example Testcases | Run | Submit